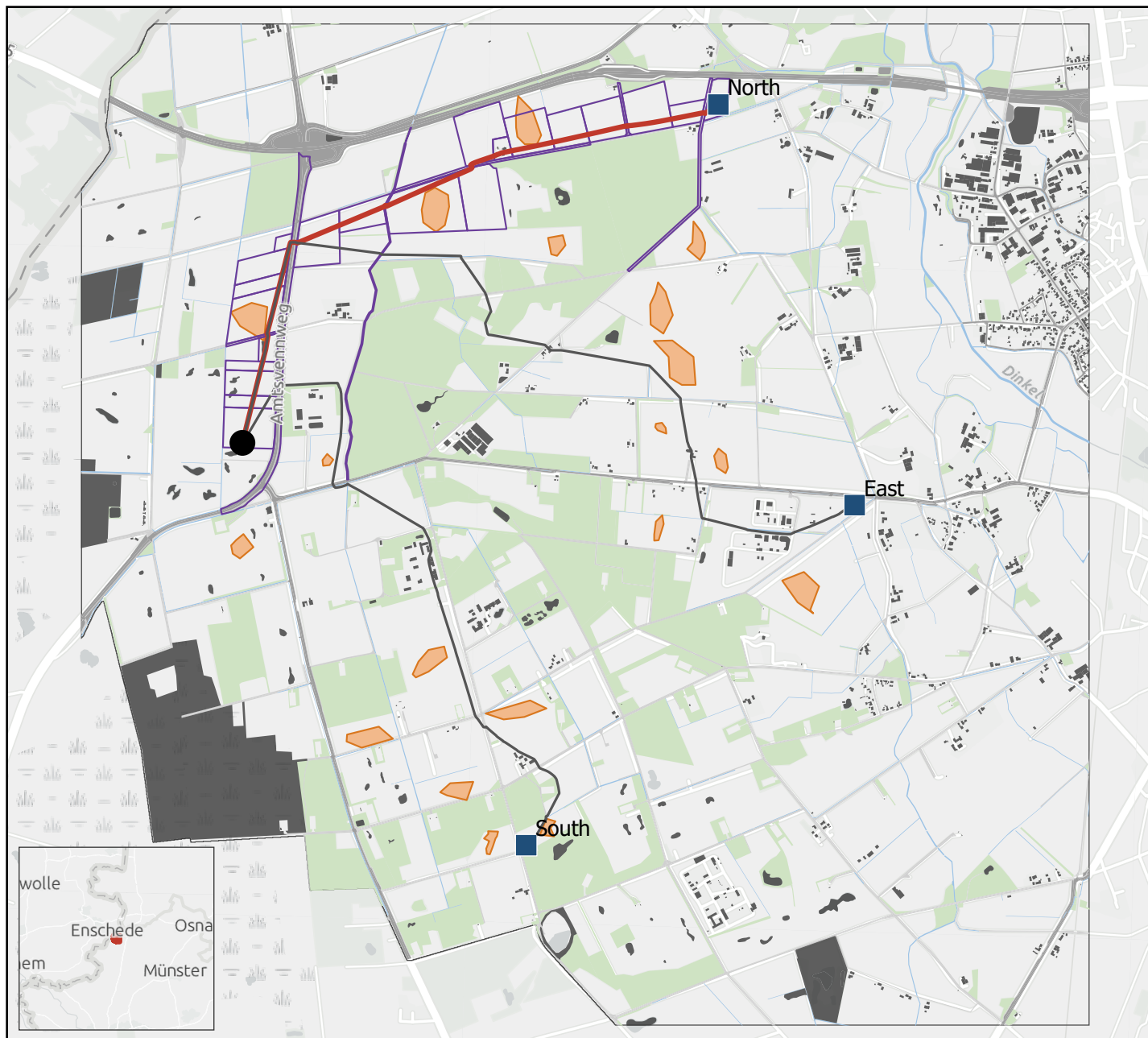


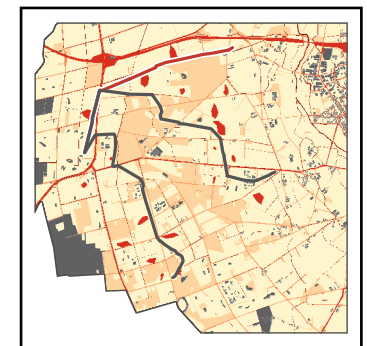


Connection point	Length (km)	Cost (€)	Average cost (€/m)
North	3.01	61,942	20.6
East	4.34	91,603	21.1
South	3.24	83,450	25.8

- Wind farm
- Grid connection points
- Least-cost route - North (€61,942, 3.0 km)
- Alternative - East (€91,603, 4.3 km)
- Alternative - South (€83,450, 3.2 km)
- Parcels crossed (28)
- Contaminated sites - €2,000/m
- No-go: buildings, standing water, bog
- Roads - €1,500/m
- Paths - €500/m
- Watercourses - €2,500/m
- Woodland and scrub - €100/m
- Study area

Cost (€/m)

- No-go (impassable)
- Open land (slope) - 3-50
- Woodland - c. 100
- Paths - c. 500
- Roads, contam. sites - 1,500-2,000
- Watercourses, overlaps - 2,500+



0 0.25 0.5 1 km

Least-cost cable route for a wind farm grid connection - Gronau-Epe (NRW)

Cost-distance analysis on a 2 m grid - Course exercise (GIS-Akademie Berlin, Spatial Analyst, Tü 4)

Data: ALKIS and DTM © Land NRW / Geobasis NRW, dl-de/zero-2-0. Wind farm, connection points and contaminated sites: exercise scenario data (GIS-Akademie Berlin). Costs are model values based on the exercise parameters, not engineering estimates. CRS: ETRS89 / UTM zone 32N (EPSG:25832). Map: Carlo Piccillo, 2026. Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, © OpenStreetMap contributors, and the GIS User Community